

CHANDRABAL C

Embedded Systems Engineer

+91 - 9751720948

✉ chandrabal2104@gmail.com

🌐 [Linkedin.com/chandrabal](https://www.linkedin.com/in/chandrabal)

ABOUT ME

Innovative Embedded Systems Engineer (2025 Graduate) with hands-on experience in ESP32, ATmega328p, and IoT development. Proficient in Embedded C and hardware design using KiCad. Proven ability to develop real-time firmware and design custom PCBs for industrial and IoT applications

TECHNICAL SKILLS

- **Microcontrollers:** ESP32, ESP8266, ATmega328p, STM32 (Learning).
- **Languages:** C, Embedded C, Python (Basics).
- **Communication Protocols:** I2C, SPI, UART, MQTT, HTTP.
- **Software Tools:** KiCad (PCB Design), Arduino IDE, VS Code (PlatformIO).
- **Hardware Skills:** Circuit Debugging, Sensor Interfacing, Soldering.

WORK EXPERIENCE

Yalamanchili Software Exports | Embedded Trainee | October 2025 - Present

- Developing firmware for ESP32 and ATmega328p based applications.
- Designing hardware schematics and 2-layer PCB layouts using KiCad.
- Performing hardware-software integration and debugging using Multimeters and Logic Analyzers.

PROJECTS

Smart Water Tank Monitoring System (ESP32, IoT)

- Developed an automated system using ESP32 and Ultrasonic sensors for high-accuracy level detection.
- Integrated ThingSpeak cloud via HTTP/MQTT for real-time remote data logging.
- Implemented low-power firmware to optimize battery efficiency for IoT deployment.

Smart Vending Machine System (ESP32, Embedded System)

- Engineered firmware for automated dispensing using PWM motor control and Relays.
- Built a user interface with 4x4 Keypad and 16x2 LCD for seamless interaction.
- Integrated feedback loops for error handling and precise product dispensing.

Smart Parking System (Embedded System)

- Developed a real-time parking slot management system using IR sensors.
- Optimized sensor polling rates to ensure high responsiveness and low power usage.
- Designed a central interface to display real-time slot availability for users.

EDUCATION

B.E – Electronics and Communication Engineering,
University College of Engineering, Anna University (2021 – 2025)
CGPA: 7.3